SOEN 341 — Software Process

CareerConnect — Generative AI Usage Log

Field	Value
Student	Xavier Lagace (40341466)
Project	CareerConnect — Job Search and Application Tracking Platform
Sprint	Sprint 2
Primary work	Scrum Master / DevOps — full-stack CI and workflow documentation
AI tool	OpenAI coding assistant (Codex through the pi interface)
Log updated	October 27, 2026

1. Purpose of this log

This document records my use of generative AI while reviewing and improving CareerConnect's continuous-integration process. AI was used to inspect the repository, explain the existing CI/CD state, propose a full-stack GitHub Actions workflow, draft documentation, and explain how the code change should move through issues, branches, pull requests, and the project board.

I did not accept the output without review. I inspected the generated workflow, compared it with the project's real npm scripts and lockfiles, cross-referenced the structure with GitHub's documentation and online Node.js workflow examples, revised the documentation in a separate pull request, and verified the result using local commands and successful GitHub Actions runs.

2. Summary of AI-assisted work

Entry	Task / purpose	Area	Decision
1	Audit the existing CI/CD and development process	Repository analysis	Accepted after verification
2	Expand backend-only automation into full-stack CI	GitHub Actions	Reviewed / Modified
3	Update the README and CI documentation	Documentation	Modified substantially
4	Connect the change to an issue, branch, PR, and project workflow	Scrum / Git workflow	Adapted and applied
5	Validate the final result	Testing / independent verification	Accepted after evidence

3. Implementation record

Work item and pull requests

- GitHub issue: #59 — Add full-stack CI and document the development workflow
- Implementation branch: `chore/task-59-ci`
- Implementation commit: `0047759` — `ci: add full stack checks and document new workflow`
- Implementation pull request: #60, linked with `Closes #59`
- Documentation branch: `chore/update-ci-doc`
- Documentation commit: `f5cdd52` — `chore(docs): update stale readme and add ci doc`
- Documentation pull request: #61

Before and after

Before	After
A workflow named `backend-tests.yml` tested only the server.	`.github/workflows/ci.yml` contains separate backend and frontend jobs.
It ran only for server/workflow changes involving `dev`.	CI runs for pull requests and pushes to `dev` and `main`, and can be started manually.
The frontend test and build scripts were not checked by GitHub Actions.	Vitest and the Vite production build are checked alongside Jest.
CI behavior was not clearly documented.	The README and `docs/CI.md` explain triggers, jobs, commands, and branch flow.

Final CI behavior

The final workflow uses Ubuntu and Node.js 24. Each job checks out the repository, installs dependencies from the appropriate lockfile using `npm ci`, and runs commands from the correct package directory.

- Backend job: `npm test -- --ci` from `server/`
- Frontend job: `npm test -- --run` and `npm run build` from `client/`
- npm caching is configured separately for `server/package-lock.json` and `client/package-lock.json`
- permissions are limited to read-only repository contents
- concurrency cancels an obsolete run when a newer commit is pushed to the same ref

The work implements continuous integration. It does not claim to implement continuous deployment because no production hosting target, deployment credentials, or persistent production storage plan was selected.

4. Detailed AI usage entries

Entry 1 — Audit the existing CI/CD setup

Task ID / title: Issue #59 preparation — repository and process audit
Purpose: Determine whether CareerConnect already had CI or CD and whether user stories/tasks were documented.
AI-suggested content: The AI inspected `.github/workflows`, both package files, the README, Sprint documentation, Git history, the public GitHub issues/project, branch state, workflow runs, and deployments. It identified one backend-only GitHub Actions workflow and no deployment workflow. It also identified that user stories and tasks existed primarily as GitHub issues and project items.
Validation: I compared the findings with the repository and GitHub interface. The original workflow contained only a server test job, and the GitHub deployments list was empty. The existing project board contained the user stories and implementation tasks.
Decision: Accepted after verification.

Reflection: AI accelerated repository discovery, but conclusions about the team's process still had to be checked against the live GitHub repository rather than inferred only from local Markdown files.
Responsible person: Xavier Lagace.

Prompt evidence:

> *"can you run me through the ci/cd in this project if any. also let me know if user stories and tasks are written out"*

Relevant response/result: The audit found CI for backend tests only, no CD, and identified the GitHub Project as the appropriate source of truth for current story/task status.

Entry 2 — Create full-stack GitHub Actions CI

Task ID / title: #59 — Add full-stack CI
Purpose: Replace the limited backend workflow with one workflow that verifies both application packages.
AI-suggested content: The AI proposed `.github/workflows/ci.yml` with `pull_request`, `push`, and `workflow_dispatch` triggers; read-only permissions; concurrency cancellation; and parallel `Backend tests` and `Frontend checks` jobs. It selected the commands by reading the actual scripts in `server/package.json` and `client/package.json`.
Validation: I reviewed the YAML line by line and checked its structure against GitHub's official workflow syntax, Node.js build/test guidance, and internet examples. I confirmed that the working directories, package-lock cache paths, Node version, and npm commands match this repository. The AI assistant also ran the same commands locally, and I reviewed the results. Finally, GitHub executed both jobs successfully on the pull request and again after merge.
Decision: Reviewed / Modified. The generated workflow was retained after my review and external cross-reference.
Reflection: A plausible YAML file is not enough. The important validation was matching it to the repository's real scripts and observing it execute successfully in GitHub Actions. Separating frontend and backend into named jobs also makes a failure easier to locate.
Responsible person: Xavier Lagace.

Prompt evidence:

> *"just make sure the github workflow is documented and if not, lets make one! like ci cd, dont document every single api thing"*

Relevant AI contribution: Drafted the full-stack workflow and explained why CI could be implemented immediately while deployment should not be invented without a real target.

Entry 3 — Update and revise CI documentation

Task ID / title: CI documentation, PRs #60 and #61
Purpose: Make the README and dedicated CI document describe the workflow accurately without duplicating every API or every project-board item.
AI-suggested content: The AI initially drafted a broader README update and `docs/CI-CD.md`, including the branch flow, local commands, CI triggers, job behavior, branch-protection guidance, and future deployment

considerations.
Validation: I reviewed the generated text against the actual workflow and then made my own documentation changes in commit `f5cdd52`. I shortened the documentation, removed sections that were broader than needed, and renamed the focused document to `docs/CI.md`.
Decision: Modified substantially.
Reflection: AI produced a complete first draft, but I decided the project needed a shorter CI-focused document. Reviewing links after renaming documentation is part of validating the final result.
Responsible person: Xavier Lagace.

Prompt evidence:

> *"can you update docs and readme if neeeded with CI/CD, just make sure any stale docs are up to date"*

> *"dont document every single api thing"*

Entry 4 — Apply the issue, branch, and pull-request workflow

Task ID / title: #59 / PR #60
Purpose: Use the CI change itself to demonstrate the team's intended development workflow.
AI-suggested content: The AI explained the sequence from an open task to an implementation branch, commit, push, pull request, CI checks, review, merge, automatic issue closure, and project-board completion. It suggested using a closing keyword in the PR description.
Validation: I created issue #59, used branch `chore/task-59-ci`, opened PR #60 into `dev`, and included `Closes #59`. GitHub merged the PR and closed the linked issue. I then used a second branch and PR #61 for my documentation revisions.
Decision: Adapted and applied.
Reflection: Linking the PR to its issue created traceability between the board item and the actual commit. CI automated technical verification, while issue acceptance and review remained human responsibilities.
Responsible person: Xavier Lagace.

Prompt evidence:

> *"i need to know how the workflow looks from editing code in a file, to pushing, pr merging, creating own branch"*

> *"which tasks do i need to change to open / how can i do that with the new code instead of manually (if possible)"*

Entry 5 — Verify the workflow with local and hosted evidence

Task ID / title: #59 validation
Purpose: Confirm that the workflow was syntactically valid and that every configured command succeeded.
AI-suggested content: The AI recommended validating locally with the exact CI commands and checking

GitHub's workflow-run results after pushing. It also distinguished CI from CD so that the documentation would not claim an unimplemented deployment process.
Validation: The local checks completed successfully:

- backend: 2 Jest suites and 29 tests passed;
- frontend: 1 Vitest file and 3 tests passed;
- frontend: Vite production build completed successfully.

GitHub Actions then provided hosted confirmation:

Run	Event/ref	Backend	Frontend	Result
39864173952	PR `chore/task-59-ci`	Passed	Passed	Success
39864444214	Push to `dev` after PR #60	Passed	Passed	Success
39896882789	PR `chore/update-ci-doc`	Passed	Passed	Success
39896928079	Push to `dev` after PR #61	Passed	Passed	Success

Decision: Accepted after independent review and execution evidence.
Reflection: Cross-referencing examples helped assess the design, but successful PR and post-merge runs were the strongest proof that the workflow was accurate for CareerConnect.
Responsible person: Xavier Lagace.

5. Human review and responsibility statement

AI acted as an implementation and explanation assistant. My responsibilities included defining the goal, deciding that the project needed full-stack CI, reviewing the generated YAML, checking external references, confirming repository-specific paths and commands, deciding which documentation to keep, revising that documentation, creating and managing the GitHub issue/branches/PRs, and accepting the result only after the local and hosted checks succeeded.

The final documentation is therefore not a verbatim AI response. Commit `f5cdd52` provides direct evidence of my revisions to the initial draft. The green GitHub Actions runs provide direct evidence that the final workflow executed as intended.

6. References and evidence

1. GitHub Docs, “Building and testing Node.js” — https://docs.github.com/en/actions/use-cases-and-examples/building-and-testing/building-and-testing-nodejs
2. GitHub Docs, “Workflow syntax for GitHub Actions” — https://docs.github.com/en/actions/reference/workflows-and-actions/workflow-syntax
3. CareerConnect workflow — `.github/workflows/ci.yml`
4. CareerConnect CI documentation — `docs/CI.md`
5. CareerConnect README — `README.md`

6. GitHub issue #59 and pull requests #60 and #61
7. GitHub Actions run records linked in Entry 5

7. Final outcome

CareerConnect now has a repeatable full-stack CI workflow that checks backend tests, frontend tests, and the production frontend build on relevant pull requests and pushes. The workflow is documented, has passed repeatedly on GitHub-hosted runners, and was introduced through the same issue/branch/PR process it is intended to protect. Continuous deployment remains outside this task's scope.